Project Title: Interpretable and Bias-Aware Detection of AI-Generated Text

CSAI411 Introduction to Natural Language Processing

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Problem Statement

Universities need reliable ways to identify AI-generated writing, but many detectors are proprietary, not fully auditable, and can falsely flag human work. Liang et al. reported a 61.3% mean false-positive rate on TOEFL essays by non-native English writers, with 97.8% of those essays flagged by at least one detector [1]. In linguistically diverse higher-education settings, unsafe detection could lead to unfair academic-integrity decisions. This project investigates whether explainable classical NLP features can detect AI-generated text while measuring and reducing false positives for learner-English writing.

Objectives

The project aims to: (1) build an interpretable AI-text detector using CSAI411 techniques such as tokenization, regular expressions, n-grams, POS tagging, TF-IDF/vector-space modelling, syntactic features, NER, and evaluation metrics; (2) compare classical models with detector baselines; (3) audit performance across domains, text lengths, and native versus learner-English writing; and (4) produce a small Gradio demo that explains predictions using feature contributions.

Dataset/Data Source

The primary dataset will be HC3, a human-versus-ChatGPT comparison corpus across multiple domains [2]. Licensing will be checked per source split, since HC3 inherits stricter source-dataset licences where applicable. GPT-wiki-intro will be used as an out-of-distribution test set because it contains Wikipedia-style human and GPT-generated introductions. For fairness testing, the project will reproduce the TOEFL essay evaluation setting from Liang et al. [1] where possible. If direct TOEFL data access is restricted, the team will use TOEFL11 or ICNALE for learner-English writing, with ICNALE native-speaker samples or another named native-English corpus used for comparison, subject to access and license terms.

Methodology

Text will be cleaned and tokenized using Python, NLTK, spaCy, and scikit-learn. Features will include word and character n-grams, TF-IDF vectors, POS-tag distributions, lexical diversity, readability, named-entity density, syntactic/dependency statistics, perplexity from a fixed reference language model such as GPT-2 small, and burstiness. The core model will be logistic regression for interpretability, with random forest or gradient boosting used only as secondary comparisons if time permits. Baselines will include a GLTR-style statistical detector [3], the GPT-2-era OpenAI RoBERTa detector baseline, and an HC3 RoBERTa detector if time permits. Evaluation will report F1-macro, AUROC, AUPR, confusion matrices, subgroup false-positive rates, FPR gaps between native and learner-English writing, and TPR at FPR <= 1%, because false positives are the main risk in academic use. Feature attribution will be analysed using logistic-regression coefficients, with SHAP added only if feasible.

Expected Results

The expected output is a working NLP pipeline and demo that classify text as human-written or AI-generated while showing which features influenced the prediction. The project should show whether classical NLP features remain competitive against transformer-based baselines and whether the model produces lower and more transparent false-positive rates for learner-English writing. The final report will include performance tables, error analysis, fairness findings, and practical recommendations for cautious use in education.

References

[1] W. Liang, M. Yuksekgonul, Y. Mao, E. Wu, and J. Zou, “GPT detectors are biased against non-native English writers,” Patterns, vol. 4, no. 7, p. 100779, 2023.

[2] B. Guo et al., “How close is ChatGPT to human experts? Comparison corpus, evaluation, and detection,” arXiv:2301.07597, 2023.

[3] S. Gehrmann, H. Strobelt, and A. M. Rush, “GLTR: Statistical detection and visualization of generated text,” Proc. 57th ACL: System Demonstrations, pp. 111-116, 2019.